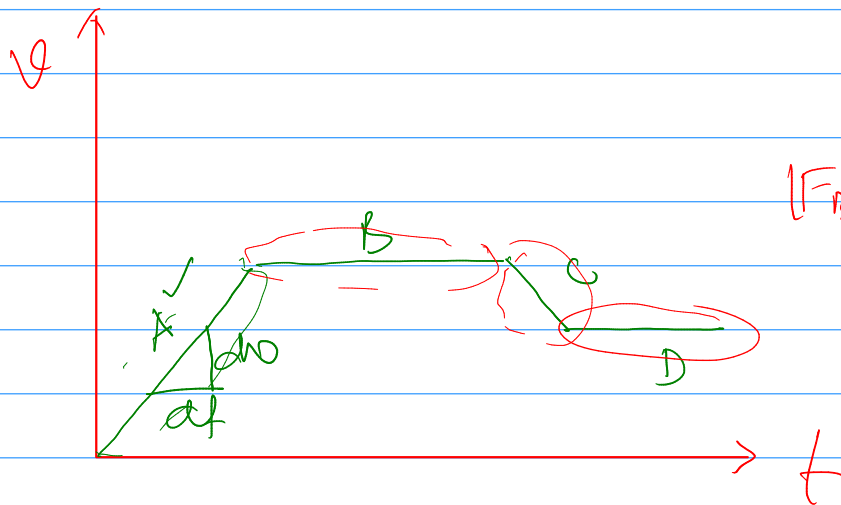


Part-01

Newton's law

Force!

long range or contact force



$$|F_B| = |F_D| = 0$$

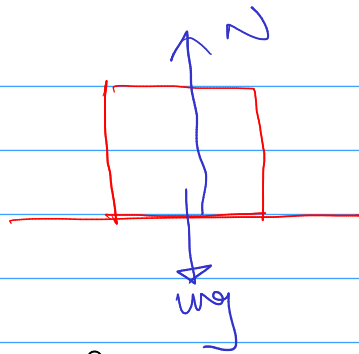
$$F_C < 0$$

$$|F_A| > 0 \quad |F_C| > 0$$

$$F = ma$$

Four Fundamental Forces!

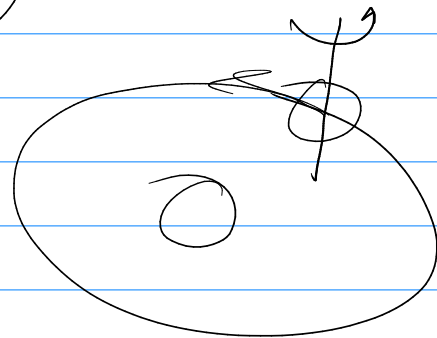
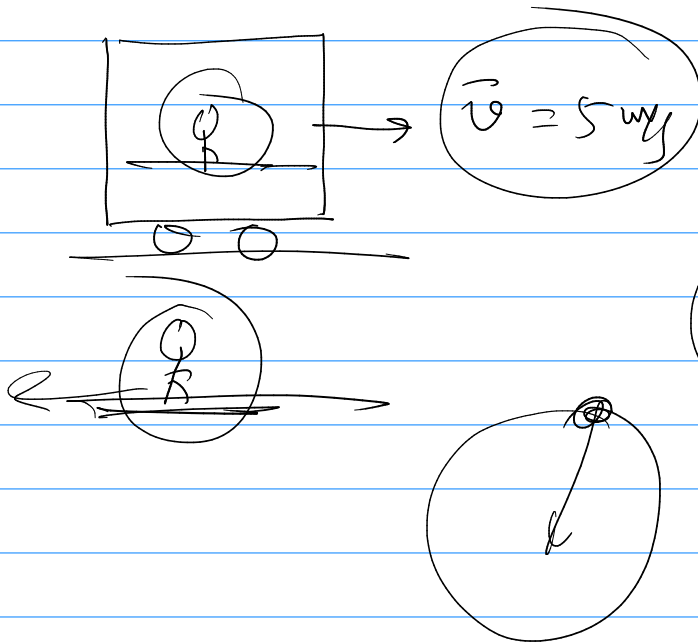
- (i) Strong force
 - (ii) Weak force
 - (iii) Gravitational force
 - (iv) Electromagnetic force
- Short range $\approx 10^{-18} \text{ m}$
- Long range $\approx \infty$



$\vec{v}_{20}$

Newton's law of Motion:-

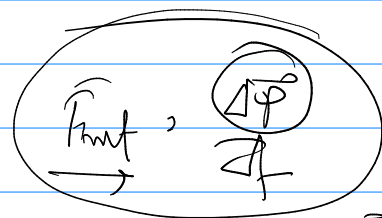
① Law of Inertia:-



$a = 0.039 \text{ m/s}^2$ ✓

Law II:-

$$\vec{F}_{\text{net}} = m\vec{a}$$



$\vec{p} = m\vec{v}$

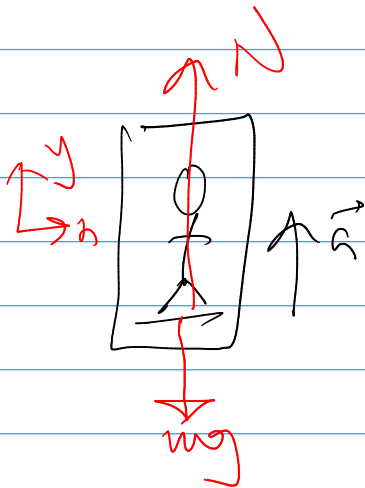
$$\frac{d\vec{p}}{dt} = m \frac{d\vec{v}}{dt} = m\vec{a} = \vec{F}$$

$$\vec{F}_{\text{net}} = \frac{d\vec{p}}{dt}$$

$$\vec{F} = m\vec{a}$$

$$\vec{p} = m\vec{v}$$

$$\frac{d\vec{p}}{dt} = m \frac{d\vec{v}}{dt} = m\vec{a} = \vec{F}_{\text{net}}$$



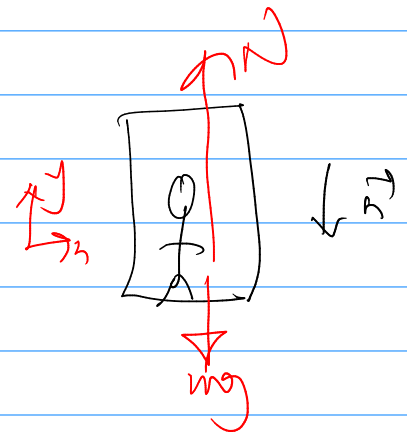
Free body diagram

$$\sum F_y = m\vec{a}$$

$$\Rightarrow N - mg = ma$$

$$\Rightarrow N = mg + ma$$

$$\Rightarrow \boxed{N = m(g + a)}$$



$$\sum F_y = m\vec{a}$$

$$\Rightarrow N - mg = -ma$$

$$\Rightarrow N = mg - ma$$

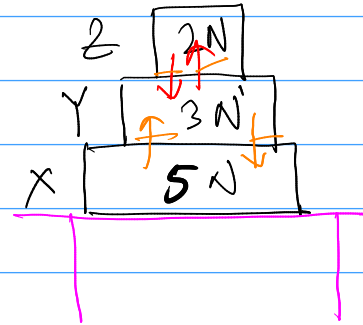
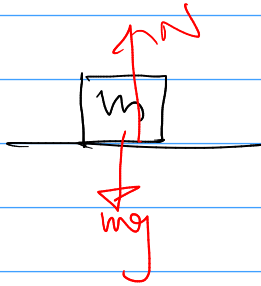
$$\Rightarrow \boxed{N = m(g - a)}$$

$$\boxed{a = g}$$

$$\boxed{N = 0}$$

Law III:

Action = - Reaction



$$F_{ZY} = 2N$$

$$F_{YT} = 2N$$

$F_{12} = \text{Force on 1 due to 2}$

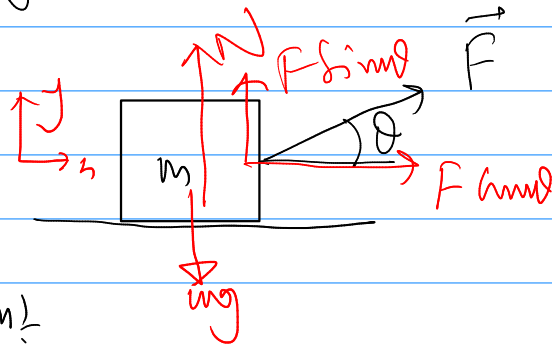
$$F_{TX} = 10N$$

T

$$F_{XT} = 10N$$

$$F_{mty} = ?$$

$$F_{XY} = 8N$$



$$a_z = ?$$

Free body diagram:

$$\sum F_y = 0$$

$$\Rightarrow N + F \sin \theta - mg = 0$$

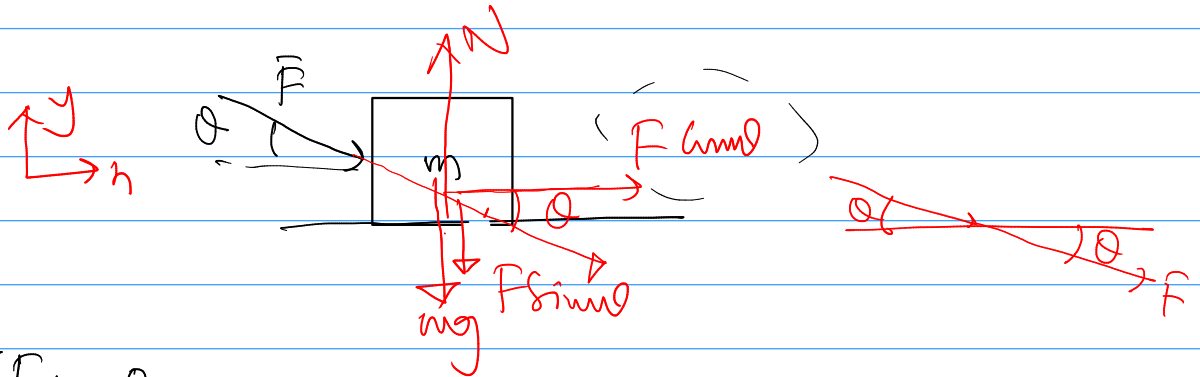
$$\Rightarrow N = mg - F \sin \theta$$

Apparent weight

$$\sum F_x = ma$$

$$\Rightarrow F_{\text{normal}} = ma$$

$$\Rightarrow \boxed{a = \frac{F_{\text{normal}}}{m}} \quad \checkmark \quad m/s \checkmark$$



$$\sum F_y = 0$$

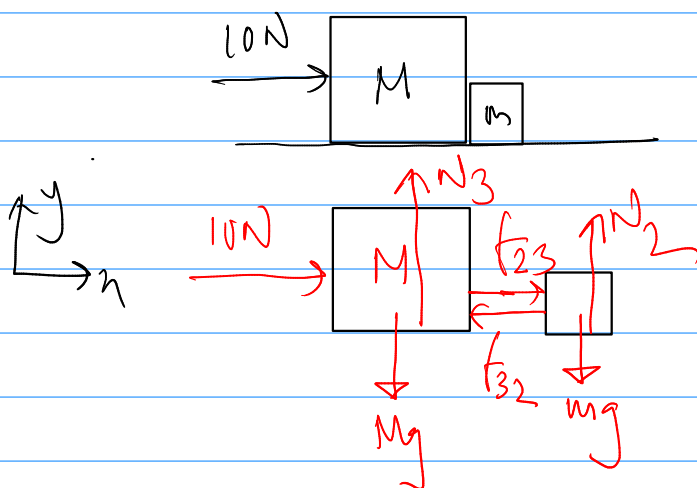
$$\Rightarrow N - mg - F \sin \theta = 0$$

$$\Rightarrow \boxed{N = mg + F \sin \theta}$$

$$\sum F_x = ma$$

$$\Rightarrow F_{\text{normal}} = ma$$

$$\Rightarrow \boxed{a = \frac{F_{\text{normal}}}{m}}$$



$$M = 3 \text{ kg}$$

$$m = 2 \text{ kg}$$

$$a = ?$$

$$\boxed{|f_{23}| = |f_{32}|}$$

2kg body:

$$\sum F_y = 0$$

$$\Rightarrow N_2 - mg = 0$$

$$\Rightarrow \boxed{N_2 = mg}$$

$$\sum F_x = ma$$

$$\Rightarrow \boxed{f_{23} = 2a} \quad \text{--- (1)}$$

3kg body:

$$\sum F_y = 0$$

$$N_3 - Mg = 0$$

$$\Rightarrow \boxed{N_3 = Mg}$$

$$\sum F_x = ma$$

$$\Rightarrow 10 = f_{32} = 3a$$

$$\Rightarrow \boxed{10 = f_{23} = 3a} \quad \text{--- (11)}$$

(1) + (11)

$$\cancel{f_{23}} = 2a$$

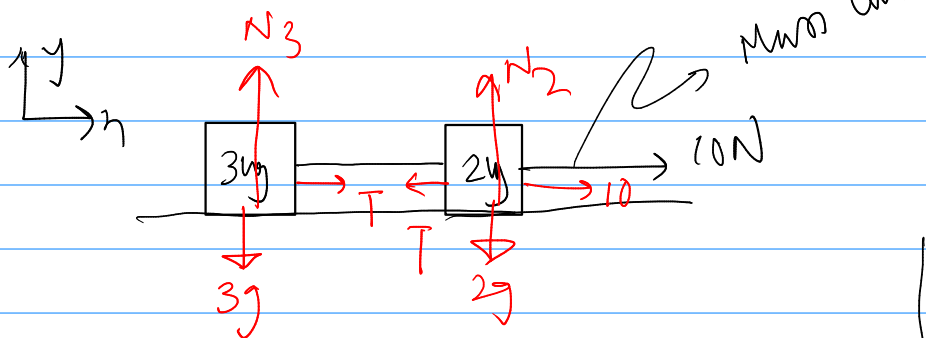
$$10 - \cancel{f_{23}} = 3a$$

$$10 = 5a$$

$$\Rightarrow \boxed{\vec{a} = 2 \text{ m/s}^2 \hat{i}}$$

$$(1) \Rightarrow f_{23} = ma = 2 \times 2 = 4 \text{ N}$$

$$10 - f_{23} = 10 - 4 = 6 \text{ N}$$



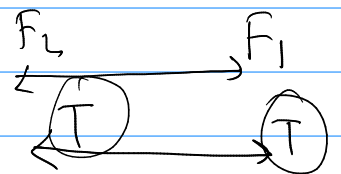
$$a = ?$$

3kg body:

$$\sum F_y = 0$$

$$\Rightarrow N_3 - 3g = 0$$

$$\Rightarrow \boxed{N_3 = 3g}$$



$$F_1 \neq F_2$$

$$F_{\text{net}} = F_1 - F_2$$

$$a = \infty$$

$$a = \frac{F}{m} = \frac{F_{\text{net}}}{0}$$

$$\sum F_n = ma$$

$$T = 3a \quad \text{--- (i)}$$

For 2kg body:

$$\sum F_y = 0$$

$$N_2 - 2g = 0$$

$$N_2 = 2g$$

$$\sum F_n = ma$$

$$10 - T = 2a \quad \text{--- (ii)}$$

$$\text{[(i) + (ii)]} \quad T = 3a$$

$$10 - T = 2a$$

$$10 = 5a$$

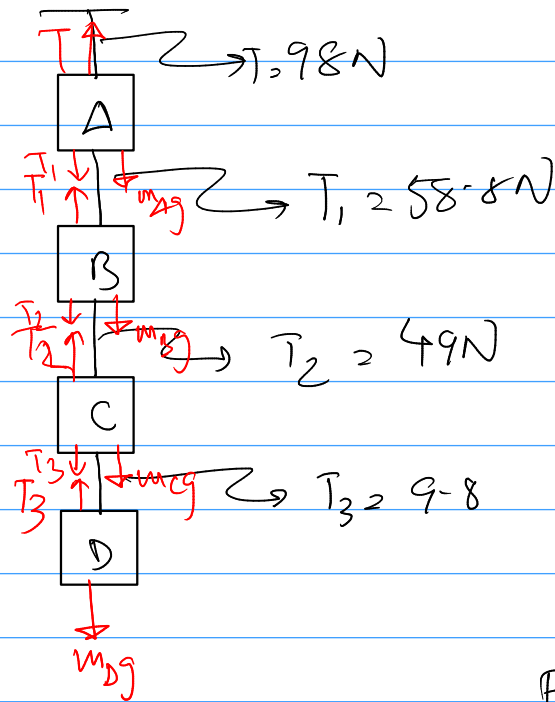
$$a = 2 \text{ m/s}^2$$

$$\text{(i) } T = 3a$$

$$T = 3 \times 2 = 6 \text{ N}$$

For 4kg body

$$18 - T = 4a$$



$$m_A = ?$$

$$m_B = ?$$

$$m_C = ?$$

$$m_D = ?$$

For D:

$$\sum F_y = 0$$

$$T_3 - m_D g = 0$$

$$\Rightarrow m_D = \frac{T_3}{g} = \frac{9.8}{9.8} = 1 \text{ kg}$$

For C:

$$\sum F_y = 0$$

$$T_2 - T_3 - m_C g = 0$$

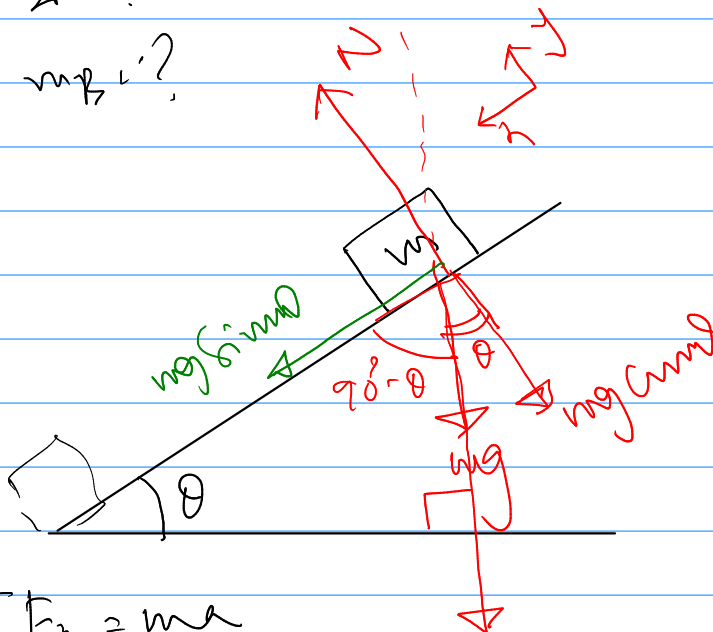
$$\Rightarrow m_C = \frac{T_2 - T_3}{g}$$

$$\Rightarrow m_C = \frac{49 - 9.8}{9.8}$$

$$\Rightarrow \boxed{m_C = 4 \text{ kg}}$$

At ω
 $m_A = ?$

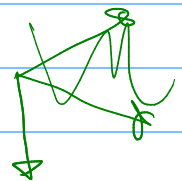
$m_B = ?$



$$\sum F_x = ma$$

$$\Rightarrow mg \sin \theta = ma$$

$$\Rightarrow \boxed{a = g \sin \theta}$$



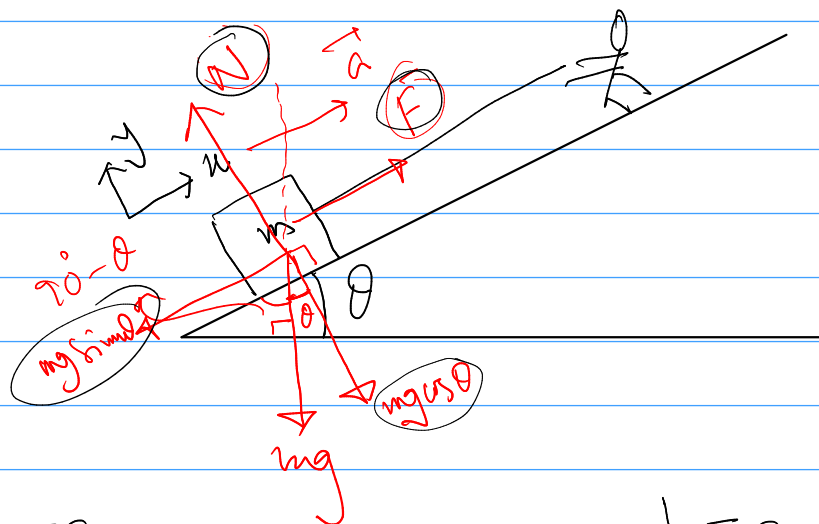
$$\sum F_y = 0$$

$$\Rightarrow N - mg \cos \theta = 0$$

$$\Rightarrow \boxed{N = mg \cos \theta}$$

Apparent weight

Part-02



$$F = ?$$

$$a = 2 \text{ m/s}^2 \checkmark$$

$$m = 150 \text{ kg}$$

$$\theta = 20^\circ$$

$$\sum F_y = 0$$

$$\Rightarrow N - mg \cos \theta = 0$$

$$\Rightarrow \boxed{N = mg \cos \theta} \rightarrow \text{Apparent weight}$$

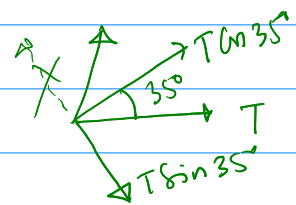
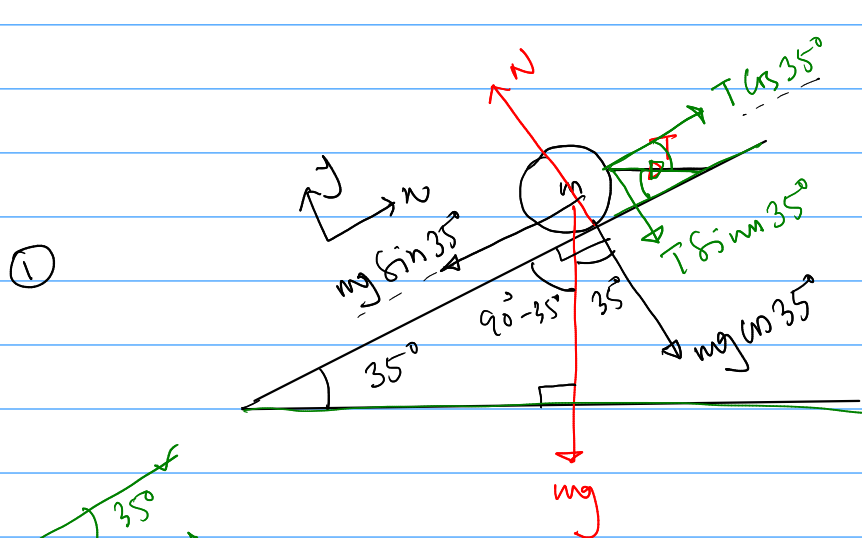
$$\sum F_x = ma$$

$$F - mg \sin \theta = ma$$

$$\Rightarrow F = mg \sin \theta + ma$$

$$\Rightarrow \boxed{F = 802.7 \text{ N}}$$

Ans



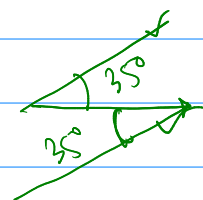
$$m = 5 \text{ kg}$$

①

① Draw the free body diagram.

② Calculate T .

③ Calculate apparent weight of the ball.



$$\text{② } \sum F_x = 0$$

$$\Rightarrow T \cos 35^\circ - mg \sin 35^\circ = 0$$

$$\Rightarrow T \cos 35^\circ = mg \sin 35^\circ$$

$$\Rightarrow T = mg \tan 35^\circ$$

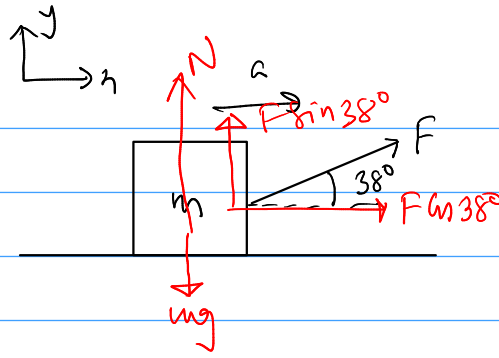
$$\Rightarrow \boxed{T = 34.31 \text{ N}} \quad \underline{\text{Ans}}$$

$$\text{③ } \sum F_y = 0$$

$$\Rightarrow N - T \sin 35^\circ - mg \cos 35^\circ = 0$$

$$\Rightarrow N = T \sin 35^\circ + mg \cos 35^\circ$$

$$\Rightarrow \boxed{N = 59.8 \text{ N}} \quad \underline{\text{Ans}}$$



$$m = 3 \text{ kg}, F = 15 \text{ N}$$

$$\textcircled{i} \quad \sum F_x = ma$$

$$\Rightarrow F \cos 38^\circ = ma$$

$$\Rightarrow a = \frac{F \cos 38^\circ}{m}$$

$$\Rightarrow \boxed{\vec{a} = 3.94 \text{ m/s}^2 \hat{i}}$$

① Calculate $\vec{a}$.

② Calculate N .

③ What is $F_{\min}$ force required to lift off the block from the surface.

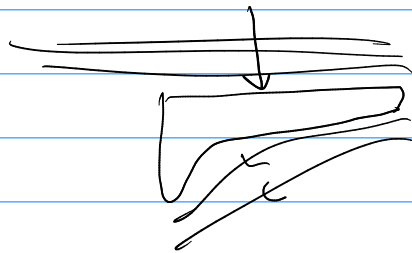
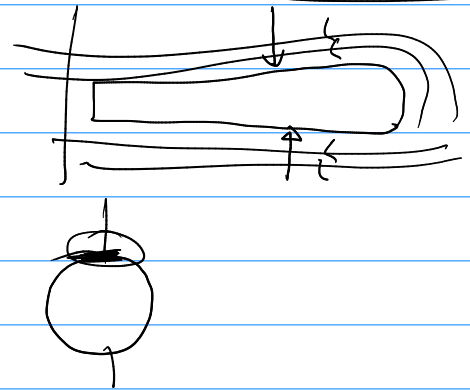
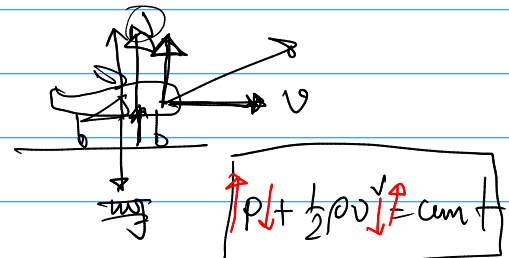
$$\textcircled{ii} \quad \sum F_y = 0$$

$$\Rightarrow N + F \sin 38^\circ - mg = 0$$

$$\Rightarrow N = mg - F \sin 38^\circ$$

$$\Rightarrow N = 20.17 \text{ N}$$

③



$$\textcircled{iii} \quad \sum F_y = 0$$

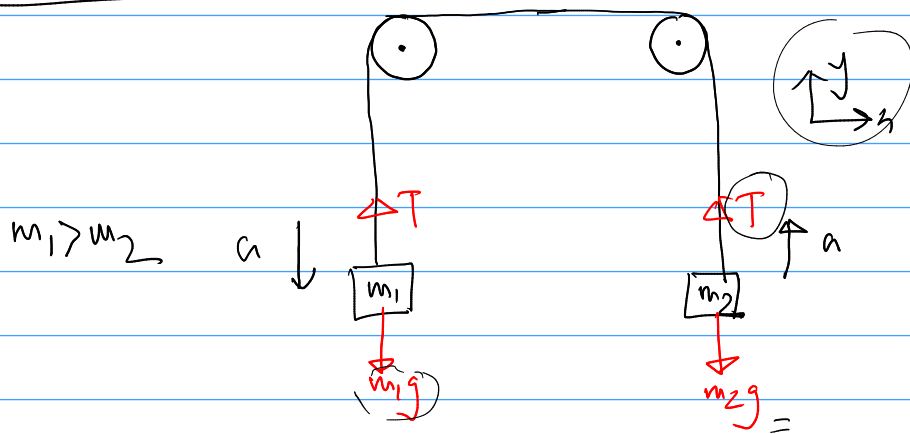
$$N + F_{\min} \sin 38^\circ - mg = 0$$

$$\Rightarrow F_{\min} \sin 38^\circ = mg$$

$$\Rightarrow \boxed{F_{\min} = \frac{mg}{\sin 38^\circ} = 47.7 \text{ N}}$$

Ans

pully system:



$$\begin{aligned} -m_1g + T_2 &= m_1a \\ T - m_2g &= m_2a \end{aligned}$$

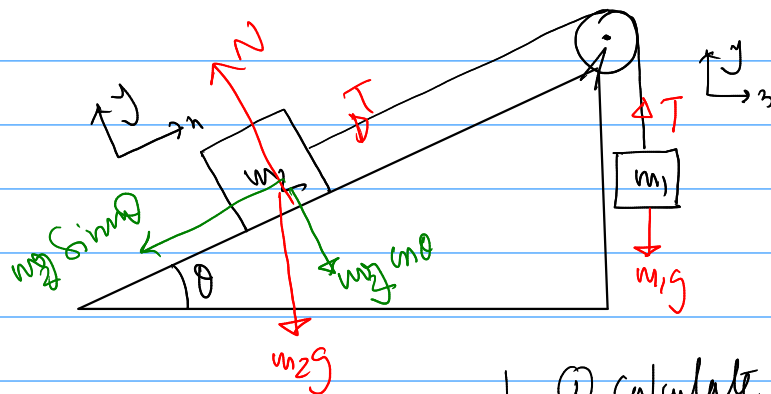
$$\sum F_{\text{net}} = ma$$

$$m_1g - T = m_1a$$

$$T - m_2g = m_2a$$

$$(m_1 - m_2)g = (m_1 + m_2)a$$

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2}$$



$$m_2 = 4 \text{ kg}$$

$$\theta = 35^\circ$$

$$m_1 = ?$$

Sch: ① For m_1 :

$$\sum F_{\text{net}} = 0$$

$$T - m_1g = 0$$

$$\therefore T = m_1g \quad \text{--- ①}$$

For m_2 :

$$\sum F_x = 0$$

$$T - m_2g \sin \theta = 0$$

$$\therefore T = m_2g \sin \theta$$

$$\therefore m_1g = m_2g \sin \theta \quad [\text{using ①}]$$

① Calculate m_1 so that the system doesn't move.

② If m_1 is 1 kg more than you have calculated in part ①, what should be $\bar{a}$.

③ If m_1 is 1 kg less, than you have calculated in part ①, what should be $\bar{a}$.

④ If the string is cut somehow what should be $\bar{a}$ of the the blocks?

$$\Rightarrow \boxed{m_1 = m_2 \sin \theta = 4 \sin 35^\circ = 2.29 \text{ kg}} \quad \underline{\text{Ans}}$$

(iv) $m_2 = 4 \text{ kg}$, $m_1 = 2.29 + 1 = 3.29 \text{ kg}$

For m_2 : $-m_2 g \sin \theta + T = -m_2 a$

$$\Rightarrow m_2 g \sin \theta - T = m_2 a \quad \text{--- (1)}$$

For m_1 : $\Sigma F_{\text{net}} = m_1 a$

$$T - m_1 g = m_1 a \quad \text{--- (11)}$$

$\{(1) + (11)\} \Rightarrow$

$$m_2 g \sin \theta - T = m_2 a$$

$$T - m_1 g = m_1 a$$

$$(m_2 \sin \theta - m_1) g = (m_1 + m_2) a$$

$$\Rightarrow a = \frac{m_2 \sin \theta - m_1}{m_1 + m_2} g$$

$$\Rightarrow \boxed{\vec{a} = -1.34 \text{ m/s}^2 \hat{i}} \quad \underline{\text{upward}}$$

(iii) $m_2 = 4 \text{ kg}$, $m_1 = 2.29 - 1 = 1.29 \text{ kg}$

$$\boxed{\vec{a} = +1.84 \text{ m/s}^2 \hat{i}} \quad \underline{\text{downward}}$$

(iv)

For m_1 : $\vec{a}_1 = -9.8 \hat{j}$

For m_2 : $\vec{a}_2 = -g \sin \theta \hat{i}$

Ans